

**WRV SERIES (250 TO 10,000 LITERS)**

# Industrial Compressed Air Receiver Tanks

Heavy-Duty Carbon & Stainless Steel Pressure Vessels Fabricated to IS 2825 & ASME Section VIII Standards for Pulsation Dampening & Bulk Moisture Separation



## **250L-10,000L**

Vertical & Horizontal  
Standard Range

## **7 TO 40 BAR**

Standard 10/16 Bar & PET  
40 Bar Vessels

## **IS 2825 / ASME**

Boiler Quality SA 516 Gr.  
70 Fabrication

## **SAFETY KIT INCLUDED**

Certified Relief Valve,  
Gauge & Drain Cock

## THERMODYNAMIC ARCHITECTURE & ENGINEERING

# Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WRV Series industrial air receiver tanks act as the vital pneumatic energy reservoir and primary moisture knockout vessel in compressed air installations. Fabricated in vertical and horizontal configurations from boiler-grade SA 516 Gr. 70 carbon steel (or SUS304 stainless steel), WRV vessels dampen pressure pulsations from reciprocating and screw compressors, satisfy sudden high-volume intermittent demand peaks, and eliminate compressor rapid cycling to save significant electrical power.

**Working Principle:** High-velocity discharge air from air compressors enters the bottom nozzle of the vertical receiver tank. The sudden drop in air velocity upon entering the large vessel volume causes heavy water droplets and oil aerosols to separate mechanically by gravity, collecting in the deep dished end for automatic removal via a zero-air-loss drain valve. The stored buffer of compressed air absorbs downstream demand spikes, stabilizing system ring-main pressure and preventing compressor motor overheating caused by frequent loading and unloading cycles.

## SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Compressor Discharge (Pulsating) → Receiver Tank Inward Tangential Nozzle → Gravitational Water Knockout → Stable High-Volume Buffer (7–16 bar) → System Discharge Header



### Boiler Quality SA 516 Grade 70 Fabrication

Constructed from high-tensile boiler-quality carbon steel plates (SA 516 Gr. 70) with dished ends formed by hot spinning. Welded via automated Submerged Arc Welding (SAW) for flawless, uniform weld root penetration.



### Full Code Compliance (IS 2825 / ASME Sec VIII)

Engineered and manufactured strictly in accordance with Indian Boiler Regulations (IS 2825 Class II/III) or ASME Section VIII Division 1, accompanied by comprehensive third-party inspection certificates.



### Complete Factory Safety Package Included

Each vessel is shipped complete with a factory-calibrated spring-loaded safety relief valve, large dial glycerin-filled pressure gauge, manual drain cock, and inspection handhole / manhole.



### Multi-Layer Anti-Corrosion Surface Treatment

Vessels undergo commercial abrasive grit blasting to SA 2.5 standard, followed by high-build epoxy primer internally/externally and durable polyurethane (PU) finish coat resistant to harsh industrial environments.



## CERTIFIED ENGINEERING RATINGS

# Technical Specifications Matrix

Full standard model range & operational parameters

**Standard Design Conditions:** Design Codes: IS 2825 Class II / ASME Sec VIII Div 1 | Working Pressures: 7.0, 10.0, 16.0, and 40.0 bar g | Hydrostatic Test Pressure: 1.5× Design Pressure | Material: SA 516 Gr. 70 Boiler Plate | Finish: SA 2.5 Grit Blast + Epoxy + PU

| MODEL CODE | CAPACITY LITERS | ORIENTATION | WORKING BAR    | SHELL DIA MM | OVERALL H/L MM | IN/OUT FLANGE   | NET WT KG |
|------------|-----------------|-------------|----------------|--------------|----------------|-----------------|-----------|
| WRV 025    | 250 L           | Vertical    | 10 bar g       | Ø 500 mm     | 1,550 mm       | 1" BSP          | 120 kg    |
| WRV 050    | 500 L           | Vertical    | 10 bar g       | Ø 650 mm     | 1,850 mm       | 1.5" BSP        | 190 kg    |
| WRV 100    | 1,000 L         | Vertical    | 10 bar g       | Ø 800 mm     | 2,250 mm       | 2" BSP / Flange | 340 kg    |
| WRV 150    | 1,500 L         | Vertical    | 10 bar g       | Ø 950 mm     | 2,450 mm       | 2" Flange       | 480 kg    |
| WRV 200    | 2,000 L         | Vertical    | 10 bar g       | Ø 1,050 mm   | 2,650 mm       | 3" Flange       | 620 kg    |
| WRV 300    | 3,000 L         | Vertical    | 10 bar g       | Ø 1,250 mm   | 2,950 mm       | 3" Flange       | 890 kg    |
| WRV 500    | 5,000 L         | Vertical    | 10 bar g       | Ø 1,500 mm   | 3,450 mm       | 4" Flange       | 1,450 kg  |
| WRV 1000   | 10,000 L        | Vertical    | 10 bar g       | Ø 1,800 mm   | 4,450 mm       | 6" Flange       | 2,950 kg  |
| WRV 050-HP | 500 L           | Vertical    | 40 bar g (PET) | Ø 600 mm     | 2,050 mm       | 1" High Press   | 380 kg    |
| WRV 100-HP | 1,000 L         | Vertical    | 40 bar g (PET) | Ø 750 mm     | 2,550 mm       | 1.5" High Press | 690 kg    |

## ENGINEERING SELECTION & OPERATIONAL NOTES

- Horizontal configurations (WRV-H) with compressor mounting top platforms manufactured to custom dimensions.
- High pressure 40 bar g models engineered specifically for PET bottle stretch blow-molding installations.
- SUS304 / SUS316L stainless steel vessels available for pharmaceutical, food processing, and semiconductor cleanrooms.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

# Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

**Manufacturing Plant Central Ring Mains** Eliminates network pressure drop and provides instantaneous compressed air storage during shift startup surges.

**PET Stretch Blow Molding (40 bar)** Buffers high-pressure air demand during rapid mold bottle blowing, preventing compressor pressure collapse.

**Pneumatic Sandblasting & Shot Peening** Maintains constant high-velocity nozzle pressure required for uniform surface cleaning without compressor stalling.

**Pneumatic Bulk Material Conveying** Powers high-velocity dense-phase air pulses to propel powders and cement grains through long transport pipes.

**Emergency Standby Air Storage** Provides critical reserve air volume to complete emergency machinery cycle shutdown during sudden electrical blackouts.

| Engineering Comparison: Win WRV vs. Uncertified Commercial Air Tanks |                                                                  |                                                              | Standard ISO 9001:2015 Quality |
|----------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------|
| Engineering Parameter                                                | Win Equipments Engineered Solution                               | Standard / Imported Benchmark                                |                                |
| Plate Material                                                       | SA 516 Gr. 70 boiler-quality plate with mill test certificates   | Commercial MS sheet with variable tensile strength           |                                |
| Welding Quality                                                      | Automated submerged arc welding (SAW) + 100% NDT inspection      | Manual MMA stick welding prone to slag inclusions & porosity |                                |
| Hydrostatic Proof                                                    | Tested to 1.5× working pressure witnessed with stamp certificate | Minimal air pneumatic bubble test (Extremely hazardous)      |                                |
| Safety Fittings                                                      | IBR-approved bronze/SS safety valve + glycerin gauge included    | Shipped as bare shell; user must source unmatched fittings   |                                |

## MANUFACTURING STANDARDS & FACTORY VERIFICATION

# Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

### ✓ Factory Acceptance Testing (FAT)

- 100% Radiographic / Ultrasonic NDT weld examination on longitudinal and circumferential seams
- 1.5× Hydrostatic proof test held for minimum 30 minutes under third-party inspection
- Dry film paint thickness (DFT) gauge verification (>120 microns total paint system)
- Safety valve popping pressure calibration and certified test bench stamp

### ★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



**Manufacturer Warranty Assurance:** 60 Months structural integrity warranty on pressure vessel shell and dished ends. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

## WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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